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Title:

**The Impact of Time Constraints on Gender
Performance Gap: Evidence from Chess Player
during COVID-19**

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Abstract:

Women, on average, receive lower wages compared to men, and one contributing factor to this wage disparity is the difference in their competitive performance. This paper investigates the impact of the COVID-19 pandemic and time pressure as stress-inducing stimuli on the performance of male and female chess players. The study's sample includes 445,941 players from the Fédération Internationale des Échecs (FIDE), representing 198 different federations, spanning the period from February 2015 to April 2023. Chess players compete in three types of games—standard, rapid, and blitz—differentiated only by time constraints. Player performance is measured through their international rating across all three formats, and analyzed using a pooled least squares model.

The findings reveal that the average international rating declined for both genders following the onset of the pandemic. However, this decline was not directly attributable to COVID-19; instead, the influx of amateur players with below-average ratings during the pandemic contributed to the overall drop in ratings. After controlling for factors such as gender, age, and virtual variables, the model demonstrates that COVID-19 had a positive effect on players' ratings, particularly in standard games, which have the least time pressure. Women, who tend to have lower average age and international ratings than men, experienced a smaller decline in ratings during the pandemic and, according to the model's results, saw a greater increase in ratings compared to men. Contrary to previous studies, no significant relationship was found between increased time pressure and a disproportionate decline in women's performance relative to men's during the pandemic.

Keywords: COVID-19, Time pressure, Gender performance difference, Competitiveness, Chess

Literature review

Generally, women are less likely to be in STEM jobs and have lower average salaries than men (Blau & Kahn, 2017). On the other hand, according to the fact that in management with a productive approach, the manager must try to create a competitive environment, therefore one of the influential factors in increasing income in job markets is the speed of employees and the ability to compete in a limited time (Bloom, Proper, Seiler & Van Reenen, 2015; Niederle & Vesterlund, 2011). To answer the question of whether gender has an effect on the ability of people to perform better under time pressure, it is possible to examine the performance of players in professional competitions. In other words, the difference in performance in sports competitions can give a good explanation for the difference in wages in STEM jobs.

The purpose of this research is to study the difference in the performance of the two genders under stressful conditions. In this study, the anxiety caused by the time limit and the Covid-19 as a secondary shock is investigated. The World Chess Federation (FIDE) regularly publishes information about the official chess tournaments held by chess players in three categories: Standard, Rapid, and Blitz. The difference between

these three types of games is the time constraint of the games; In this case, the performance of the players can be checked under the condition of anxiety due to the time limit. The study of chess players is important in several ways: first, the data is extracted from an environment where the participants are highly experienced and skilled and want to participate in competitions. Second, chess is an environment in which players receive a great deal of feedback and information about the relative abilities of their opponents, as players can analyze themselves and their opponents' games with computers. Third, FIDE data is at three different time levels and in this sense, it is like a special laboratory condition in which only the performance speed of the players is evaluated under time limit conditions. Fourth, we study gender interaction effects in an environment where men and women compete on equal conditions; A condition in which chance has the least impact (Grabner, 2014).

As a result, examining the difference in performance between the two genders in competitive conditions along with anxieties such as time constraints and COVID-19 can be an answer to the question of whether there is a logical reason for the difference between the income and job positions of the two genders.

The epidemic has had a greater anxiety effect on women than men, because women are often expected to take care of the elder person of the family and children in critical times, and this effort to balance work and care responsibilities can lead to job burnout and increased anxiety levels. This study is important because the time limit in our decisions and choices is an inseparable part of the present era. Due to the advancement of technology and as a result of the increase in the speed of information transmission in the world, people are forced to make better decisions and react more appropriately in the shortest time to be successful (Dilmaghani, 2019). This research is trying to investigate more dimensions of the difference between the performance of women and men in the conditions of competition, under time pressure and anxiety. This article is important because it can measure the difference in the performance of women and men and is a reason for the low percentage of women in management jobs compared to men. On the other hand, on average, the income difference between the two genders is significant (Blau Vekan, 2017). Investigating COVID-19 as a double anxiety shock can highlight these gender differences more so that policymakers and employers have more appropriate expectations from their employees, create a healthier work environment, and determine fairer wages.

Descriptive Statistics

The study of the performance of chess players is related to the study of gender and has several important features that make it similar to competitive job opportunities. First, chess is like STEM jobs in which women are less frequently present, so according to the data of FIDE organization in April 2023, not one woman is ranked among the top 100 players of the World Chess Federation. Also, there has never been a female world chess champion in history. Second, there has generally been a negative view of female players at the top level in the past. According to the data published by the FIDE website in February 2022, only 11% of the players participating in mixed tournaments were women, and only 2% of the chess grandmasters were women. Time pressure is an inseparable part of chess, although only the time and number of participants do not reflect the level of competitiveness of the players. Competitiveness is assumed to be reflected in the number of moves recorded for a chess game. In particular, while in popular culture the game of chess is associated with the concept of "checkmate", checkmate rarely happens on the chessboard among professional players. Professional players usually quit before checkmate, which means they accept defeat.

The data used in this research includes three general categories of chess players of the World Chess Federation. These three types of games are: standard, rapid, and blitz the only difference between them is the time limit. Blitz games last between 3 and 10 minutes, speed games usually last between 10 minutes

and an hour, and standard games can last for hours. All data can be accessed through the World Chess Federation database.

The data of all games in all three types (standard, fast, blitz) and the compilation of these data are explained using Table 1. Note that international ratings of players below 1,000 are not reported by the World Chess Federation. The data of this research is monthly and will continue from February 2015 until April 2023, which is the last month before the end of Covid-19.

World Chess Federation data includes an identification number, name, age, gender, federation, number of games played in each month, month and year, titles, international rating, and activeness of the player. The identification number of the players is unique. According to FIDE rules, if a person has more than one identification number, he must declare it. Like other sports, in chess, the federation usually means the name of the players' country, but in some cases, the players can play under the flag of their country without the federation. Due to the separation between games, the type of game column was also added to the data, and the dummy variable of COVID-19 was also added to the column of all data. According to the data, COVID-19 starts in March 2020 and ends on May 5th, 2023; Therefore, after March 2020, this column will be one, and before that, it will be zero. Some observations were removed due to lack of name, gender, and federation, and players who had changed their gender and federation were also removed from the data for consistency. The number of removed data is less than one percent of the data. Also, all the months in which the players did not play were removed because FIDE publishes the information of all the players every month, even the players who did not play. The players do not participate due to two reasons, one is the need to pay cash for the matches registered by the World Federation. Also, the players attend the most training to ensure an increase in the international fide rating. Therefore, a large amount of observations were only repetitions of previous international degrees, which were removed.

Table 1 shows the statistical description of chess players. The number of women players is much less than men, and only ten percent of the total observations are made up of women. After cleaning the data, the total number of participants in all three types is 445,941, each of which has participated in at least one type of game. In general, women have a lower average age than men and the standard deviation of their age is also lower. The average age for women is 19 years and their standard deviation is only 13 years, which means that 95% of female players are between the ages of 6 and 32. Whereas, men have an average age of 32.5 years and the standard deviation of their age is 19.5 years, which means that 95% of men are between 13 and 52 years old. This age difference shows that men not only generally start participating in chess tournaments later, but also they continue playing until a greater age. This amount of age difference is very significant.

By comparing the number of observations in different parts of Table 1, it can be seen that the number of participants in the standard game is more than in the rapid and blitz games so the blitz game with 169,895 participants includes only about half of the players of the standard game. Rapid game with 247,289 participants includes 72% of standard game players and the number of standard players is 341,470. Players participating in games with higher speed are younger on average. The minimum age of players in both genders is almost the same and is about 3 to 5 years. and there is no significant difference, but the maximum age for men is 122 years and for women is 97 years.

A) Aggregated Games

Mean (Number)	Standard deviation or percentage	Maximum	Minimum
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Fide rating				
All	1698.01	358.8	2,986	1,000
Male	1720.40	354.43	2,986	1,000
Female	1511.58	339.11	2,686	1,000
Age				
All	31.14	19.36	122	3
Male	32.45	19.47	122	3
Female	19.24	13.39	97	3
Observation				
All	445,941			
Male	401,788	90%		
Female	44,153	10%		

B) Games Standard				
	Mean (Number)	Standard deviation or percentage	Maximum	Minimum
Fide rating				
All	1727.34	354.15	2,882	1,001
Male	1748.96	348.22	2,882	1,001
Female	1535.11	348.38	2,686	1,001
Age				
All	31.23	19.51	118	4
Male	32.67	19.63	118	4
Female	19.07	13.24	97	5
Observation				

All	341,470	
Male	305,503	90%
Female	35,967	10%

C) Rapid Games				
	Mean (Number)	Standard deviation or percentage	Maximum	Minimum
Fide rating				
All	1605.84	355.15	2,919	1,000
Male	1627.57	354.84	2,919	1,000
Female	1443.77	313.04	2,645	1,000
Age				
All	29.32	19.02	101	3
Male	30.61	19.19	101	3
Female	18.36	13.20	96	3
Observation				
All	247,289			
Male	221,132	89%		
Female	26,157	11%		

D) Games Blitz				
	Mean (Number)	Standard deviation or percentage	Maximum	Minimum
Fide rating				
All	1706.67	361.44	2,986	1,000
Male	1731.49	359.27	2,986	1,000
Female	1528.47	325.05	2,676	1,000

Age

All	28.55	18.36	122	4
Male	29.87	18.56	122	4
Female	17.94	12.35	96	5

Observation

All	169,895	
Male	151,062	88.91%
Female	18,833	11.09%

Table 1. Statistical description of chess players in different games of the World Chess Federation (February 2015 - April 2023)

Fide Rating calculation

The FIDE Rating system is a numerical system in which fractional scores are converted to rating differences and vice versa. Its function is to produce measurement information of the best statistical quality. The international rating of each player is a natural number, the higher it is, the more powerful that player is.

This number is determined based on the results of official matches between players. In each match, the winner gets certain points and the same amount of points is deducted from the loser. The amount of this point depends on the international rating difference between the two players before the match.

In determining the international rating of players, factors such as the previous rating, the result of the game, and the coefficient K, which is known as the "development coefficient", have an effect. This coefficient, which is one of the numbers 10, 20, or 40, is determined according to the age and international rating of the players. The lower the age and international rating of the players, the bigger this coefficient is. Also, this number is 40 for players who are not rated. In fact, this coefficient makes the international rating of players with higher international rating, age, and playing less affected by the results of their last games.

$$R_{\text{new}} = R_{\text{old}} + K * (s - E)$$

- R_{new} : the new rating of the players.
- R_{old} : A rating before the match.
- K : development coefficient
- S : Result of the match (losse=0, win = 1, draw= 0.5).

E : Expect of winning

$$E = \frac{1}{10 \frac{(R_{\text{old}} - R_{\text{opponent}})}{400} + 1}$$

In order to have a more accurate understanding of international degrees, it is better to check them. The chess.com website, which is the most famous and popular online chess website, uses the information of the American Chess Federation to divide players into different classes in terms of skill level as follows:

- International rating 1,000-1,200: Class A Players New player
- International rating 1,200-1,399: Class D players usually beginners
- International rating 1,400-1,599 Class C players average player
- International rating 1,600-1,799: player B above average
- International rating 1,800-1,999: Class A player Strong player of the club

FIDE also considers titles for professional players, one of the conditions of which is to obtain these international degrees. An international rating of 2,000 is the start of the professional world of chess, which is a necessary condition for getting the title of master candidate. Also, an international rating of 1,400 indicates the average level of chess skill. For example, Lee Chess, one of the most popular free chess websites, registers its new members with a default rating of 1,500, which is the average rating of the website. It is also stated in Table 1 that the international rating of men in all three types of competitions is between 1,600 and 1,799, which according to the explanations, is above average. The international rating of women in all three types of competitions is numerically between 1,400 and 1,599, which according to the explanations, is the average of the club or competition. The maximum international rating of men in each standard game is 2,882, which is the lowest maximum value among the three types of matches. The highest number is the men's international blitz with 2,986. The maximum international rating of women in all three types of games is around 2,660. In fact, in all types of games, the international degree required to get the degree of grandmaster has been obtained by the best male and female players. This international degree difference between the maximum international degree of women and men shows that the number of male grandmasters is much more than the number of female grandmasters.

COVID-19

On December 12, 2019, a group of patients in China's Hubei Province, in the city of Wuhan, began experiencing symptoms of unusual pneumonia. This illness did not respond well to standard treatments. On December 31, the WHO Office in China was informed of several cases of patients with unspecified symptoms including shortness of breath and fever in Wuhan, China. All primary cases appear to be linked to the Huanan Seafood Wholesale Market (Centers for Disease Control and Prevention). On March 11, 2020, the World Health Organization declared COVID-19 as a pandemic after more than 118,000 cases of COVID-19 were identified in 114 countries and 4,291 deaths were confirmed.

In 2020, chess was affected by the pandemic like any other sport, matches were postponed or canceled and classes were temporarily closed. However, the nature of chess helped players, fans, and enthusiasts stay active and engaged during this crisis, and online chess experienced an unprecedented boom. For example, in May 2020, FIDE launched a massive online project called "Checkmate Coronavirus". A marathon of 720 hours of non-stop gameplay and 2,762 matches played by 120,000 unique participants from at least 140 countries.

The rapid spread of COVID-19 globally has been met with an extraordinary range of government responses, including school closures, travel restrictions, bans on public gatherings, emergency investments in healthcare facilities, new forms of social welfare provision, contact tracing, and Other interventions to prevent the spread of the virus, increase the health of systems and manage the economic consequences of

these measures (Hale et al., 2021). Finally, in May 2023, most countries have lifted their policies related to COVID-19 and the World Health Organization has declared the end of the Public Health Emergency of International Concern.

Data and methodology

Based on the statistical values of Table 2, the international level of players has generally decreased after the COVID-19 pandemic. In standard games, the lowest amount of decrease is seen, to know the significance of the difference between the average international rating before and during COVID-19, a t-test was taken which was big enough in all types to show the difference is significant.

Another point is that in rapid and blitz games, where the playing time is less and the players have to play at a higher speed, this international rating difference increases. Contrary to the review literature, the international rating of women in rapid games is lower than all types of games, even blitz games which are the fastest type of game. Also, the statistical variables indicate that men have been affected by COVID-19 more than women.

	Before Covid-19	During Covid-19	Difference	Mean difference test T-test [P-value]
Aggregate games				
All	1723.03 (365.51)	1644.56 (357.47)	85.00	236.02 [0.000]
Male	1533.49 (341.15)	1464.53 (329.81)	68.96	72.66 [0.000]
Female	1745.85 (351.49)	1666.09 (354.57)	85.72	228.98 [0.000]
Standard games				
All	1742.10 (352.83)	1692.30 (354.82)	49.80	111.97 [0.000]
Male	1548.54 (348.63)	1502.17 (345.57)	46.37	36.75 [0.000]

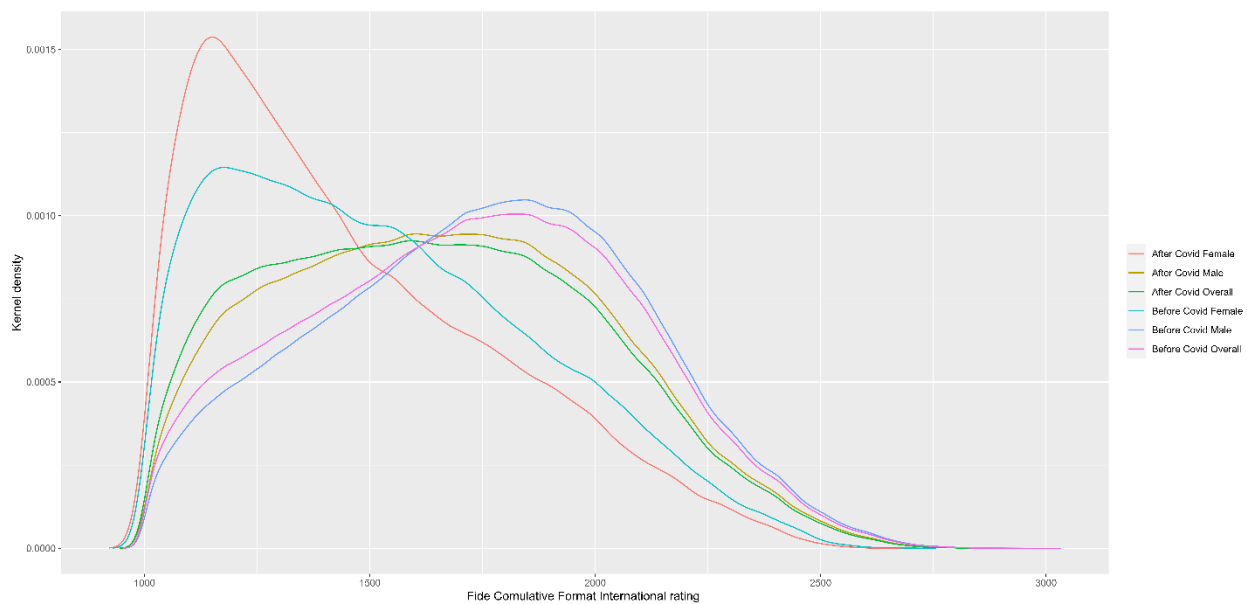
Female	1746.10 (346.53)	1713.14 (349.59)	50.96	118.33 [0.000]
Rapid games				
All	1647.07 (356.32)	1538.11 (341.73)	108.96	167.70 [0.000]
Male	1474.88 (319.26)	1390.21 (294.43)	84.67	49.802 [0.000]
Female	1760.19 (354.70)	1554.48 (342.96)	115.71	162.68 [0.000]
Blitz game				
All	1744.92 (360.31)	1635.94 (352.76)	108.98	119.25 [0.000]
Male	1562.26 (328.35)	1467.05 (309.66)	95.21	41.02 [0.000]
Female	1770.19 (357.23)	1659.76 (351.96)	110.43	113.68 [0.000]

Note: The numbers in the parentheses indicate the standard error. The numbers in brackets indicate the p-value

By examining graph A-1, it can be seen that the kernel density of the international rating of female players with an international rating below 1,400 has increased a lot after COVID-19 compared to before, this means that the proportion of semi-professional players with an international rating below 1,400 Compared to international degrees, it has increased a lot. The highest kernel density difference in international rating is 1,200. For men, this increase in kernel density continues up to the international rating of 1,600, which is the highest kernel density difference in the international rating of 1,200. But this difference is less in women than in men. In fact, more new player-arrived women have entered the sample than men which might be a result of the queen gambit mini-series which the main role was a woman who becomes an international chess champion for the first time in a male-dominated chess environment. Chess has also grown and the number of observations in the game in the last month of April 2023 is 413,945, which is about 2 times the first month of February 2015, which was 200,033. The same trend can be seen in rapid and blitz games. The number of observations in the game of the last month of rapid is 261,622, which is about 6 times that of the first month of data (45,310). In the game, the last month of blitz is 182,879, which is about 5 times

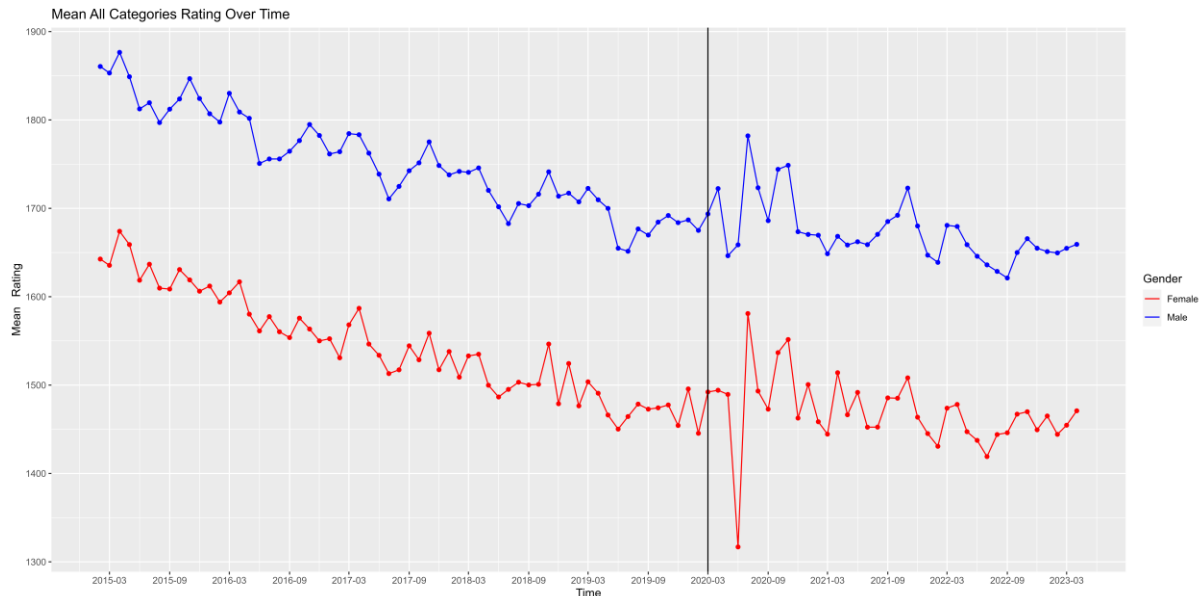
the first month of data (37,071). In fact, the number of chess enthusiasts has increased, and these people who are newcomers often have an international rating below 1,700.

On the other hand, for women with an international degree above 1,400, the density of the international degree after COVID-19 is much lower than before COVID-19, this difference decreases as the international degree increases. In men, with an international rating above 1,600, this decreasing trend can be seen similarly. The reason for the decreasing trend is several factors; The first reason is that there are fewer professional people, which is completely normal, and the second reason is the low impact of factors such as COVID-19 on the performance of these professional players, after the international rating of 2,500, which is the international rating for the great masters, the number of these people is very small, and these people are the most professional players. continuously In all the graphs of Appendix A, this distinction between the density of players with an international rating above 2,500 is almost zero(the other kernel density graphs are available in [Kernel-density](#)).



By examining the monthly trend of changes in the average international rating, it can be seen that graph 1 shows the average monthly international rating separately for men and women in all games. In this diagram, because the maximum amount of participating players and the maximum amount of matches are for standard games, diagram 1 is similar to diagram 2. The downward trend of the average international ratings from the observations at the beginning to the end of the study period is significant. This decreasing trend is completely natural due to the increase in the number of players of the World Federation. The reason for this is that new players who are interested in playing chess have increased and another reason is that the desire of players to have an international rating has increased. Previously, most of the players of the World Chess Federation were more professional players. But with chess becoming more popular and joining the federation among the common people, this process intensified. In the month of June 2020 (vertical line in the graph), an abnormally sharp decrease in international rating among female players can be seen. In the months following the publication of the news of the confirmation of COVID-19 as a pandemic by the World Health Organization, a large number of games were closed and postponed, due to which instability arose in these months. The amount of monthly fluctuations has a greater effect on rapid and blitz games. The reason for this is the smaller number of players in these games, which causes the variance of the average scores of the players to be higher and they are more affected by the monthly effects. As we can see, the highest

amount of fluctuations is included in blitz games which have the least number of participating players (the other kernel density graphs are available in [Mean rating](#)).



Graph 1: the average rating of the players in all categories aggregated

Note: the vertical line indicates the start of the COVID-19

Model

In the following, we will examine the model used in this research. According to Graph 1, the international rating changes in different months of the year, so there is a possibility of the month and year affecting the results of the average international ratings. This happened because of the holding of some important competitions such as Grand Cherix FIDE, and World Cup in those months of the year. These competitions and the motivation to participate in such competitions will make stronger players participate in these competitions, and this will increase the average international rank in the months of holding. For this reason, the variables of year and month are get as factor in the model to control the seasonality. Players are expected to become more professional due to more practice. Players need more time to practice and see more courses to progress, so the age of the players is also controlled and it is expected that the skill and therefore the international rating of the players will increase with age.

As it is clear in tables 1 and 2 and graphs, significant differences can be seen between the international degree of men and women. This difference exists in all months, because of this issue, being a woman is another control variable. After pandemic, many changes took place, such as closing and postponing games, forcing people to wear masks, holding online competitions, keeping distance between people, and anxiety caused by the presence of COVID-19 in chess. In Table 2, this difference between the average international rank of players before and after Corona is clearly visible. This difference has a range of about 50 to 115 international degrees according to the type of competition and gender, so the presence of the impact of Covid-19 as a test variable is added to the model so that its coefficient answers the research question. Finally, the variable "COVID-19 $\times$ female" should also be controlled because the performance of women and men is different in this anxiety situation. In fact, this control variable is the most important variable and the subject of discussion whether women's performance in competitions has decreased compared to men or not. Due to the existence of three standard, rapid and blitz game categories, the model is used for all types of game categories. For this, we have taken standard matches as a reference and compared other matches.

Because the other two types of games are more similar to each other from the point of view of time pressure and in standard games time plays a lesser role and the performance of the players reflects the skill level of the people rather than their ability to control time and anxiety caused by lack of time, also The World Federation awards titles based on standard matches.

The regression model (1) are specified as follows:

$$\text{Rating}_{it} = \alpha + \gamma_1 \text{Game}_i + \gamma_2 \text{COVID19}_t + \gamma_3 \text{Female}_i + \gamma_4 \text{Age}_{it} \\ \beta \times \text{COVID19}_t \times \text{Female}_i + \text{Year}_t + \text{Month}_t + e_{it}$$

Model (1) is repeated for three other types of games because Rating_{it} is regressed for the same specific type of game. In equation (1), the variable Rating_{it} is the international rank of player i in month t . Game_i specifies the type of games, which is one of the three types of standard, rapid, and blitz, and is defined as a factor in R . The parameter γ_1 is the coefficient of Game_i , which shows how much the type of speed of playing the game changes the international rating of the players. COVID19_t is the dummy variable of COVID-19, which is one means during the period of Covid-19 and zero means before Covid-19, the parameter γ_2 is the coefficient of COVID19_t , which shows how players change their performance due to the COVID-19 anxiety. Female_i is a dummy variable of gender, one means female, and zero means male, parameter γ_3 is the coefficient of Female_i , which shows how many degrees being female changes the international rating of the players. Players who have changed their gender during the data have been removed for consistency. Age_{it} is the age variable of player i in each month t , and the parameter γ_4 is the coefficient of Age_{it} , which shows that increasing the age by one month is a change in the international rating of the players. β is the coefficient of the test variable $\text{COVID19}_t \times \text{Female}_i$, which shows how much a woman in the conditions of COVID-19 causes a change in the international rating of the players. The parameter α is the intercept from the origin of the model and e_{it} is the part of the disturbance of the model caused by unobserved factors.

The results of model (1) for four aggregated games, standard, rapid, and blitz are reported in Table 3. All estimates are estimated using the difference in differences method. According to diagram 1, the effect of seasonality is noticeable, so the dummies of year and month have been controlled. As mentioned in Table 1, women have a lower international rating than men, so being a woman is expected to reduce the amount of international rating. The second row of Table 3 shows that the international rank in all types of games for women is less than men above 100 rating. This means that in standard games, women have an average of -145 degrees less international degree than men, which is significant at the one percent level, in the

same way, it is -111 degrees in the rapid game and -134 degrees in blitz, so there is a relationship between There is no increase in time pressure and no further decrease in the international rank of women.

The coefficient of our experiment is " $\text{COVID19}_t \times \text{Female}_i$ " and in all cases, it has increased the international rank of women compared to men; In fact, the shock caused by COVID-19 has not only caused a decrease in the international rank of women, but their international rank has increased in all types of competitions compared to men all coefficient was significant.

Independent variable fide rating	All Categories	Standard	Rapid	Blitz
COVID-19*Female	4.338*** (1.003)	1.560 (1.310)	14.504*** (1.921)	5.232* (2.598)
Female	-136.120*** (0.576)	-145.820*** (0.724)	-111.589*** (1.180)	-134.257*** (1.564)
COVID-19	24.272*** (1.162)	26.201*** (1.447)	25.940*** (2.437)	22.100*** (3.230)
Age	4.702*** (0.007)	4.067*** (0.009)	5.844*** (0.015)	5.890*** (0.020)
Rapid (Reference Standard)	-84.733*** (0.361)			
Blitz (Reference Standard)	17.267*** (0.451)			
Year Dummies	Controlled	Controlled	Controlled	Controlled
Month Dummies	Controlled	Controlled	Controlled	Controlled
Observations	5,312,987	3,492,416	1,162,258	664,051

Table 3. Estimation results of difference in differences model (February 2015 - April 2023)

Note: The symbols ***, **, * indicate the significance of the coefficients at the levels of 99.99%, 99%, and 95%, respectively.

The numbers in parentheses indicate the standard error.

As can be seen in the "COVID19_t × Female_i" row, the highest increase in international rank among rapid players is 14 international rank for each female player, which is significant at the 1% level, followed by this rate in power games. Asa has increased by 5 degrees, which is significant at the five percent level, and finally by 1 degree among standard players, which is not even significant as the p-value indicate. Therefore, it can be concluded that women perform better than men in chess game with increasing time pressure and anxiety.

As the age of people increases, their international rank increases by about five to six ranks in each type of game every month. Although according to Table 1, the age range of the players is from 4 years to over 100 years, it is expected that this rate of increase in the international level is not constant during the life of the players. The rate of increase in the international rank of younger players with lower international ranks is faster than players with higher international ranks, and as the age increases and the international ranks rise, the growth of the players' international rank should decrease.

In general, the international rating of players has increased in the ranks of Covid-19. This amount of increase in standard games is the highest and is equal to 26 degrees, and it decreases in rapid and blitz games and reaches 25 and 22 degrees, respectively. Considering the decrease in the average international rank of both sexes after the occurrence of covid-19, controlling for other conditions, covid-19 not only did not decrease the international rank of players but also increased it. Also, at the end of the rapid games compared to the standard games, there were 84 points less in general and this amount was the opposite in the rapid games and the performance of the players increased by 17 international degrees.

Result

According to the results of this study, which was obtained from the review of 5,312,987 observations of chess players of the World Federation on a monthly basis; Chess players in the covid-19 era have a lower international rank on average than before covid-19. In the comparison between the games, the average international rating of the players in the rapid and blitz games has decreased more than the standard matches after the occurrence of COVID-19. Not to mention that the players of blitz games have the lowest average age in the three types of matches. This age difference is more among men.

The amount of international rating for women has decreased on average less than men. However, the results of the model show that COVID-19 has caused an increase in the international ranking of players and this increase is significant in all types of games at the level of five percent, even for women this increase was more than for men. Whereas, according to previous studies, this amount of difference was expected to be more in favor of men.

During the COVID-19 era, as announced by FIDE, FIDE's online chess tournaments have expanded and more chess classes have been held online, which helps chess players improve their skills in a less gender-dominated environment. Probably, these things have caused progress and as a result the international rating of players, especially women, has increased.

Also, the downward trend of the average international rank of chess players shows that the downward trend of the average international rank of chess players had started before COVID-19. The main reason for the decline in the international rank of chess players is the increase in the number of players of the World Chess Federation. This increase in the number of players is due to two reasons: First, the number of players interested in chess has increased. Second, the World Chess Federation is more famous and important among chess players than before. The increase in the number of chess players will undoubtedly decrease the average international rank of chess players because new players generally have less experience in official tournaments and less knowledge than older players.

Data from the FIDE website shows that the increase in the number of players was very significant, so in the first month of the study period, there were 200,033 players in the standard game, while in the last month, this number reached 413,945; In rapid games, from 45,310 players to 261,622, and in blitz games, this number has increased from 37,071 to 182,879. As it is evident, the most number of new players have experienced the rapid game and then blitz. The comparison of graphs 1 to 4 also shows that the decrease in the average international grade of fast games is more than blitz and standard.

According to the kernel density graphs in the [Git Hub](#), it can be said that among female chess players with an international rank between 1,500 and 2,000, there was the largest decrease in participation. But among men, this decrease in the participation rate is from the international level of 1,700 to 2,200. In fact, the decrease in the average international rank seems to be due to less participation of average players and a huge increase of new players. The number of professional players has not changed much, but with the decrease in the participation of the average players who make up the majority and the increase of newcomers, players with an international rank below 1,400 have decreased the average international rank.

Also, according to [graphs \(all samples, standard, rapid, blitz\)](#), it can be said that women generally have a weaker performance than men, this weaker performance is not just because of their fewer numbers. The characteristics of women show that on average women are younger and the standard deviation of their age is less than men, in fact, women start chess at a younger age and give up chess in a shorter period of time than men. Parents and coaches think that young female players have less potential than young male players; They also believe that success in chess requires inherent capacities (Arnold et al, 2023).

Even before 2023, the rules of chess were written from a male point of view male pronouns were used 179 times, and female pronouns were used only twice, in the new rule these numbers are 178 and 177 have been changed in order (Fide Rules Commission, 2023). Therefore, it can be said that one of the reasons for the early abandonment of chess by women is the lack of motivation given to them.

Finally, the remarkable thing is that women in the days of COVID-19, contrary to expectations, grew internationally in all types of games. Despite the fact that the number of new women players increased more than men which should decrease their rating. Moreover, the absence of a relationship between the increase in time pressure and the decrease in women's performance, it is not possible to give a definite answer that women show weaker performance in games with less time.

The results of the difference in differences model were in contradiction with the average international degrees that had decreased during the Corona period. One of the reasons for the decrease in performance was the sharp increase in the number of players due to the broadcast of a series called Queen Gambit, which scored 8.5 on Netflix; It also caused an increase in the sales of chess books and, as a result, a greater increase in the number of female players than male players. The main role of this series was a woman who was able to become a world champion in a patriarchal environment.